

Aditi Ghosh

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EDUCATION

- RCC Institute of Information Technology** Kolkata, India
B.Tech in Computer Science (CSE); CGPA: 8.84 / 10.0 Sep 2023 – Present
Coursework: DSA, OOP, OS, DBMS, Computer Networks, Software Engineering.

SKILLS SUMMARY

- Languages:** Python, Java, C++, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3
- Frameworks:** React.js, Next.js, Django, Node.js, Tailwind CSS, TensorFlow, OpenCV
- Tools:** Git, Docker, Azure, PostgreSQL, Firebase, Postman, Vercel
- Core Concepts:** DSA, System Design, REST APIs, Computer Vision, CI/CD

EXPERIENCE

- PayZoll** Remote
Web Developer Intern Feb 2025 – Apr 2025
 - Identified performance bottlenecks in the legacy interface utilizing older templates that hindered user retention
 - Targeted the specific need to modernize the frontend architecture for better type safety and maintainability.
 - Migrated template components written in **Octo SDK** into a **Next.js** environment using **TypeScript**, maintaining proper typing.
 - Reduced load times by **40%** and increased Google Lighthouse scores to **95+**, significantly improving **Quality Assurance** metrics.

PROJECTS

- HerShield — Women Safety WebApp** | *Next.js, Python NLP* [Code] [Live]
 - Identified the limitation of manual panic buttons in safety apps, necessitating automated text-based distress detection.
 - Targeted the need for intelligent parsing of user messages to trigger alerts without explicit user interaction.
 - Developed an NLP-driven engine using **TensorFlow** and integrated **Google Maps API** for dynamic safe-route calculation.
 - Attained **87% recall** in distress signal detection and optimized frontend performance to a **95+ Lighthouse score**.
- Beacon — Secure Communication** | *Next.js, Firebase* [Code] [Live]
 - Addressed enterprise security vulnerabilities regarding sensitive document transmission and potential data leakage.
 - Focused on establishing a "zero-trust" delivery mechanism for confidential internal files.
 - Engineered a secure delivery system featuring **AES-256 encryption**, **PIN-based MFA**, and a real-time telemetry dashboard.
 - Ensured **100% data privacy compliance** and enforced strict access boundaries through **Role-Based Access Control (RBAC)**.
- BitShred — Data Sanitization GUI** | *Next.js, TypeScript, Python* [Code] [Live]
 - Identified critical security risks associated with standard file deletion, where data remains recoverable by forensic tools.
 - Tasked with developing a military-grade data sanitization tool to ensure permanent, irreversible data destruction across partitions.
 - Engineered a **Next.js** and **Python** GUI that detects system partitions and implements cryptographic data overwriting algorithms.
 - Guaranteed **0% data recoverability** against regular data recovery tools and generated detailed audit reports for user security.

HONORS AND AWARDS

- Grand Finalist** – Smart Bengal Hackathon 2025 (Team Leader of one of the Top 30 teams out of 500+ participants)
- Status Code 1** – Team Lead for decentralized voting platform project - 2024
- Top 20 Contributor** – JWOC Winter of Code (Ranked out of 100+ participants) - 2024

LEADERSHIP & VOLUNTEER EXPERIENCE

- GDG RCCIIT** Kolkata
Web Development Lead – Organized technical workshops & hackathons. Present
- ACM RCCIIT** Kolkata
Web Development Lead Present